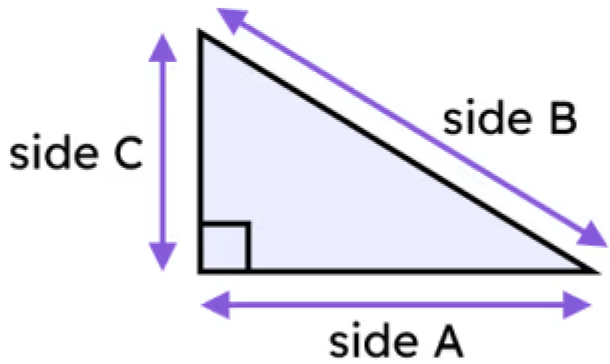
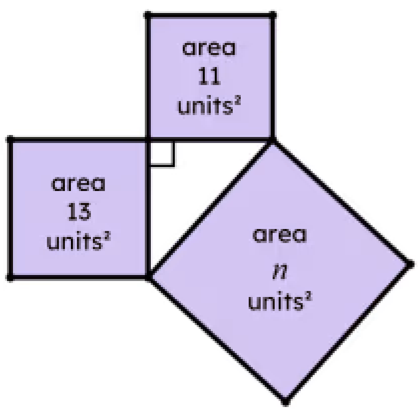
Length of the hypotenuse

1 Which of these sides is a hypotenuse? Tick **1** correct answer



- [ ] side A
- [x] side B
- [ ] side C
- [ ] none of them are hypotenuses
- [ ] it is impossible to tell

2 The triangle formed from these three squares is right-angled. What is the value of n, where n units² is the area of a square. Tick **1** correct answer

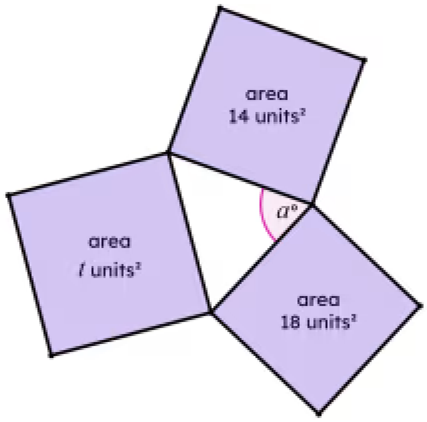


- [ ] 2
- [ ] 4.9
- [ ] 12
- [x] 24
- [ ] 290

3 If three squares with different areas are joined at their vertices, what type of triangle would be formed? Tick **1** correct answer

- [ ] acute triangle
- [x] scalene triangle
- [ ] isosceles triangle
- [ ] equilateral triangle

4 Angle $a°$ is an acute angle, but is the largest angle in this triangle. Which of these are possible values of t, where t units² is the area of a square. Tick **2** correct answers



- [ ] 17
- [x] 19
- [x] 20
- [ ] 32
- [ ] 34

5 Which of these are true for the Pythagoras' theorem? Tick **3** correct answers

- [ ] Describes a property of only equilateral triangles.
- [x] Describes a property of only right-angled triangles.
- [x] Describes a relationship between the squares of the side lengths of a triangle.
- [ ] Describes a relationship between the interior angles of a triangle.
- [x] Describes a relationship between squares formed from the sides of a triangle.

6 A right-angled triangle is formed from three squares. The area of two of the squares are 75 units² and 25 units². What are the possible areas of the third square? Tick **2** correct answers

- [ ] 3 units²
- [ ] 25 units²
- [x] 50 units²
- [x] 100 units²
- [ ] 6250 units²